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H2 Mathematics (9758)
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2026 H2 Math Vectors I Test: Solutions with comments

Total marks: 42

Full sourced questions. Give exact answers unless otherwise stated.

1 With reference to the origin O, the points A and B have coordinates $(1, 2, 0)$ and $(-5, 1, 0)$.

(a) Find the angle AOB. [3]

The vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are projected along the direction of $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ to form the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$ respectively.

(b) Find the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$. [3]

(c) Find the exact area of the triangle $O'A'B'$. [2]

[DHS 2023 Promo Q5]

Solution		Comments
(a) **[3]**	$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} -5 \\ 1 \\ 0 \end{pmatrix}$. Let $\theta = \angle AOB$. $\overrightarrow{OA} \cdot \overrightarrow{OB} = \lvert\overrightarrow{OA}\rvert \lvert\overrightarrow{OB}\rvert \cos\theta$ $\cos\theta = \dfrac{\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 1 \\ 0 \end{pmatrix}}{\sqrt{1^2+2^2}\sqrt{5^2+1^2}}$ $= \dfrac{-5+2}{\sqrt{5}\sqrt{26}} = -\dfrac{3}{\sqrt{5}\sqrt{26}}$ $\theta = 105.3^\circ$.	

(b)
[3]

$$\overrightarrow{O'A'} = \frac{\overrightarrow{OA}\cdot\begin{pmatrix}1\\2\\2\end{pmatrix}}{\left|\begin{pmatrix}1\\2\\2\end{pmatrix}\right|}\frac{\begin{pmatrix}1\\2\\2\end{pmatrix}}{\left|\begin{pmatrix}1\\2\\2\end{pmatrix}\right|}$$

$$= \frac{\begin{pmatrix}1\\2\\0\end{pmatrix}\cdot\begin{pmatrix}1\\2\\2\end{pmatrix}}{(\sqrt{1^2+2^2+2^2})^2}\begin{pmatrix}1\\2\\2\end{pmatrix} = \frac{5}{9}\begin{pmatrix}1\\2\\2\end{pmatrix}$$

$$\overrightarrow{O'B'} = \frac{\overrightarrow{OB}\cdot\begin{pmatrix}2\\1\\2\end{pmatrix}}{\left|\begin{pmatrix}2\\1\\2\end{pmatrix}\right|}\frac{\begin{pmatrix}2\\1\\2\end{pmatrix}}{\left|\begin{pmatrix}2\\1\\2\end{pmatrix}\right|}$$

$$= \frac{\begin{pmatrix}-5\\1\\0\end{pmatrix}\cdot\begin{pmatrix}2\\1\\2\end{pmatrix}}{(\sqrt{2^2+1^2+2^2})^2}\begin{pmatrix}2\\1\\2\end{pmatrix} = -\begin{pmatrix}2\\1\\2\end{pmatrix}.$$

(c)
[2]

$$\text{Area of triangle } O'A'B' = \frac{1}{2}|\overrightarrow{O'A'}\times\overrightarrow{O'B'}|$$

$$= \frac{5}{18}\left|\begin{pmatrix}1\\2\\2\end{pmatrix}\times\begin{pmatrix}2\\1\\2\end{pmatrix}\right| \qquad (|\mathbf{a}\times(-\mathbf{b})| = |\mathbf{a}\times\mathbf{b}|)$$

$$= \frac{5}{18}\left|\begin{pmatrix}2\\2\\-3\end{pmatrix}\right|$$

$$= \frac{5}{18}\sqrt{17}\text{ units}^2.$$

2 The position vectors of points A, B and C, relative to the origin O are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a}=3\mathbf{i}+2\mathbf{j}-\mathbf{k}, \quad \mathbf{b}=4\mathbf{i}-3\mathbf{j}+2\mathbf{k}, \quad \mathbf{c}=3\mathbf{i}-\mathbf{j}-\mathbf{k}.$$

(a) Find $\mathbf{a}\times\mathbf{b}$ and deduce the exact area of triangle OAB. [3]

(b) Hence, find the volume of the tetrahedron $OABC$.
[The volume of a tetrahedron is $\frac{1}{3}\times$base area $\times$ perpendicular height.] [3]

(c) Find the perpendicular distance from B to line OA. [2]

[NYJC 2024 Promo Q5]

Solution		Comments
(a) **[3]**	$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}3\\2\\-1\end{pmatrix}\times\begin{pmatrix}4\\-3\\2\end{pmatrix}=\begin{pmatrix}1\\-10\\-17\end{pmatrix}$ Area of $\triangle AOB=\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ $=\frac{1}{2}\left\lvert\begin{pmatrix}1\\-10\\-17\end{pmatrix}\right\rvert$ $=\frac{1}{2}\sqrt{1+100+289}=\frac{1}{2}\sqrt{390}$ units^2.	
(b) **[3]**	Perpendicular height of tetrahedron $=\frac{1}{\sqrt{1+100+289}}\left\lvert\overrightarrow{OC}\cdot\begin{pmatrix}1\\-10\\-17\end{pmatrix}\right\rvert$ $=\frac{1}{\sqrt{390}}\left\lvert\begin{pmatrix}3\\-1\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\-10\\-17\end{pmatrix}\right\rvert$ $=\frac{30}{\sqrt{390}}$ Volume of tetrahedron $OABC$ $=\frac{1}{3}\left(\frac{\sqrt{390}}{2}\right)\left(\frac{30}{\sqrt{390}}\right)$ $=5$ units^3.	
(c) **[2]**	Perpendicular distance from B to line $OA=\lvert\mathbf{b}\times\hat{\mathbf{a}}\rvert$ $=\frac{\sqrt{390}}{\sqrt{9+4+1}}$ $=\sqrt{\frac{195}{7}}$ or 5.28.	

3 Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC:AB=1:3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the area of triangle OAC is given by $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be determined. [2]

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}$. [3]

[MI 2024 Promo Q4]

Solution		**Comments**
(i) **[2]**	$AC:AB=1:3 \quad\Rightarrow\quad AC:CB=1:2$ $\mathbf{c}=\dfrac{(1)\mathbf{b}+(2)\mathbf{a}}{1+2}=\dfrac{2\mathbf{a}+\mathbf{b}}{3}.$	
(ii) **[2]**	Area of triangle $OAC=\dfrac{1}{3}$ Area of triangle OAB $=\dfrac{1}{3}\times\dfrac{1}{2}\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$ $=\dfrac{1}{6}\lvert\mathbf{a}\times\mathbf{b}\rvert, \qquad k=\dfrac{1}{6}.$	
(iii) **[3]**	$(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}=(2\mathbf{a}-\mathbf{b})\cdot\left(\dfrac{2\mathbf{a}+\mathbf{b}}{3}\right)$ $=\dfrac{1}{3}(2\mathbf{a}-\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$ $=\dfrac{1}{3}(4\mathbf{a}\cdot\mathbf{a}+2\mathbf{a}\cdot\mathbf{b}-2\mathbf{b}\cdot\mathbf{a}-\mathbf{b}\cdot\mathbf{b})$ $=\dfrac{1}{3}(4\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2)$ $=\dfrac{1}{3}(4(1)^2-(1)^2)$ since $\mathbf{a}$, $\mathbf{b}$ are unit vectors $=1.$	

4 **(a)** The points A, B and C have position vectors $\begin{pmatrix}1\\3\\-7\end{pmatrix}$, $\begin{pmatrix}5\\1\\1\end{pmatrix}$ and $\begin{pmatrix}-9\\8\\-27\end{pmatrix}$, with respect to the origin O.

(i) Show that the points A, B and C are collinear. [2]

(ii) Find the exact area of the triangle OAB. [3]

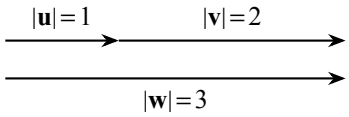
(b) The vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are defined such that $|\mathbf{u}|=1$, $|\mathbf{v}|=2$ and $|\mathbf{w}|=3$.

(i) Suppose that $\mathbf{u}+\mathbf{v}=\mathbf{w}$. With the aid of a diagram, or otherwise, explain why $\mathbf{u}$ is a unit vector that is parallel to $\mathbf{w}$. [2]

(ii) Given instead that each of the three vectors is perpendicular to the sum of the other two vectors, find the exact value of $|\mathbf{u}+\mathbf{v}+\mathbf{w}|$. [4]

[TMJC 2024 Promo Q6]

Solution		Comments
(a)(i) **[2]**	$\overrightarrow{AB}=\begin{pmatrix}5\\1\\1\end{pmatrix}-\begin{pmatrix}1\\3\\-7\end{pmatrix}=\begin{pmatrix}4\\-2\\8\end{pmatrix}$ $\overrightarrow{AC}=\begin{pmatrix}-9\\8\\-27\end{pmatrix}-\begin{pmatrix}1\\3\\-7\end{pmatrix}=\begin{pmatrix}-10\\5\\-20\end{pmatrix}$ $=-\frac{5}{2}\begin{pmatrix}4\\-2\\8\end{pmatrix}=-\frac{5}{2}\overrightarrow{AB}.$ Since $\overrightarrow{AB}$ is parallel to $\overrightarrow{AC}$, with a common point A, the points A, B and C are collinear. (Shown)	
(a)(ii) **[3]**	Area of triangle $OAB=\frac{1}{2}\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$ $=\frac{1}{2}\left\lvert\begin{pmatrix}1\\3\\-7\end{pmatrix}\times\begin{pmatrix}5\\1\\1\end{pmatrix}\right\rvert$ $=\frac{1}{2}\left\lvert\begin{pmatrix}10\\-36\\-14\end{pmatrix}\right\rvert=\sqrt{398}.$	

(b)(i) **[2]**	Since $\mathbf{u}+\mathbf{v}=\mathbf{w}$, and $\lvert\mathbf{u}\rvert=1$, $\lvert\mathbf{v}\rvert=2$ and $\lvert\mathbf{w}\rvert=3$, $\mathbf{u}$ and $\mathbf{v}$ must both be parallel to $\mathbf{w}$, as seen in the diagram.  Since $\lvert\mathbf{u}\rvert=1$, $\mathbf{u}$ is a unit vector parallel to $\mathbf{w}$.	
(b)(ii) **[4]**	Since each of the three vectors is perpendicular to the sum of the other two vectors, $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=0, \quad \mathbf{v}\cdot(\mathbf{u}+\mathbf{w})=0, \quad \mathbf{w}\cdot(\mathbf{v}+\mathbf{u})=0$ $\lvert\mathbf{u}+\mathbf{v}+\mathbf{w}\rvert^2=(\mathbf{u}+\mathbf{v}+\mathbf{w})\cdot(\mathbf{u}+\mathbf{v}+\mathbf{w})$ $=\mathbf{u}\cdot\mathbf{u}+\mathbf{v}\cdot\mathbf{v}+\mathbf{w}\cdot\mathbf{w}+\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})+\mathbf{v}\cdot(\mathbf{u}+\mathbf{w})+\mathbf{w}\cdot(\mathbf{v}+\mathbf{u})$ $=\lvert\mathbf{u}\rvert^2+\lvert\mathbf{v}\rvert^2+\lvert\mathbf{w}\rvert^2$ $=1^2+2^2+3^2=14.$ Since $\lvert\mathbf{u}+\mathbf{v}+\mathbf{w}\rvert>0$, $\lvert\mathbf{u}+\mathbf{v}+\mathbf{w}\rvert=\sqrt{14}$.	

5 Relative to the origin O, the points A and C have position vectors $\mathbf{a}$ and $2\mathbf{b}$ such that

$$\mathbf{a}=5p\mathbf{i}-2p\mathbf{j}+4p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k},$$

where p is a positive constant. It is given that $|\mathbf{a}|=2|\mathbf{b}|$.

(a) Find the exact value of p. [2]

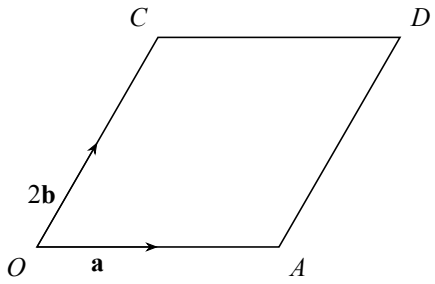
(b) Evaluate $(\mathbf{a}+2\mathbf{b})\cdot(\mathbf{a}-2\mathbf{b})$. [1]

(c) Evaluate $\mathbf{a}\times\mathbf{b}$ and hence find the area of triangle OAC. [3]

(d) Use a geometrical reason to explain why $|\mathbf{a}\times\mathbf{b}|=\frac{1}{4}|\mathbf{a}+2\mathbf{b}||\mathbf{a}-2\mathbf{b}|$. [2]

[RI 2025 Prelim P1 Q3]

Solution		Comments
(a) **[2]**	Since $\|\mathbf{a}\|=2\|\mathbf{b}\|$, $\sqrt{(5p)^2+(-2p)^2+(4p)^2}=2\sqrt{1^2+(-2)^2+2^2}$ $3\sqrt{5}p=6 \qquad (p>0)$ $p=\frac{2}{\sqrt{5}}=\frac{2\sqrt{5}}{5}.$	
(b) **[1]**	$(\mathbf{a}+2\mathbf{b})\cdot(\mathbf{a}-2\mathbf{b})=\mathbf{a}\cdot\mathbf{a}-2\mathbf{a}\cdot\mathbf{b}+2\mathbf{b}\cdot\mathbf{a}-4\mathbf{b}\cdot\mathbf{b}$ $=\|\mathbf{a}\|^2-4\|\mathbf{b}\|^2 \qquad (\|\mathbf{a}\|=2\|\mathbf{b}\|)$ $=0.$	
(c) **[3]**	$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}5p\\-2p\\4p\end{pmatrix}\times\begin{pmatrix}1\\-2\\2\end{pmatrix}$ $=p\begin{pmatrix}5\\-2\\4\end{pmatrix}\times\begin{pmatrix}1\\-2\\2\end{pmatrix}$ $=p\begin{pmatrix}4\\-6\\-8\end{pmatrix}=\frac{4\sqrt{5}}{5}\begin{pmatrix}2\\-3\\-4\end{pmatrix}$ Area of triangle $OAC=\frac{1}{2}\|\mathbf{a}\times2\mathbf{b}\|=\|\mathbf{a}\times\mathbf{b}\|$ $=\frac{4\sqrt{5}}{5}\sqrt{2^2+(-3)^2+(-4)^2}=\frac{4\sqrt{145}}{5}.$	

(d) **[2]**	Observe that $\overrightarrow{OD}=\mathbf{a}+2\mathbf{b}$ and $\overrightarrow{CA}=\mathbf{a}-2\mathbf{b}$. From (b), OD is perpendicular to CA. In addition, CA and OD, which are the diagonals of the parallelogram $OADC$, bisect each other.  From (c), area of triangle $OAC=\lvert\mathbf{a}\times\mathbf{b}\rvert$. But area of triangle OAC is also $\frac{1}{2}(CA)\left(\frac{1}{2}OD\right)=\frac{1}{4}\lvert\mathbf{a}-2\mathbf{b}\rvert\lvert\mathbf{a}+2\mathbf{b}\rvert$. Hence $\lvert\mathbf{a}\times\mathbf{b}\rvert=\frac{1}{4}\lvert\mathbf{a}+2\mathbf{b}\rvert\lvert\mathbf{a}-2\mathbf{b}\rvert$.	

END OF SOLUTIONS